YOGI VEMANA UNIVERSITY

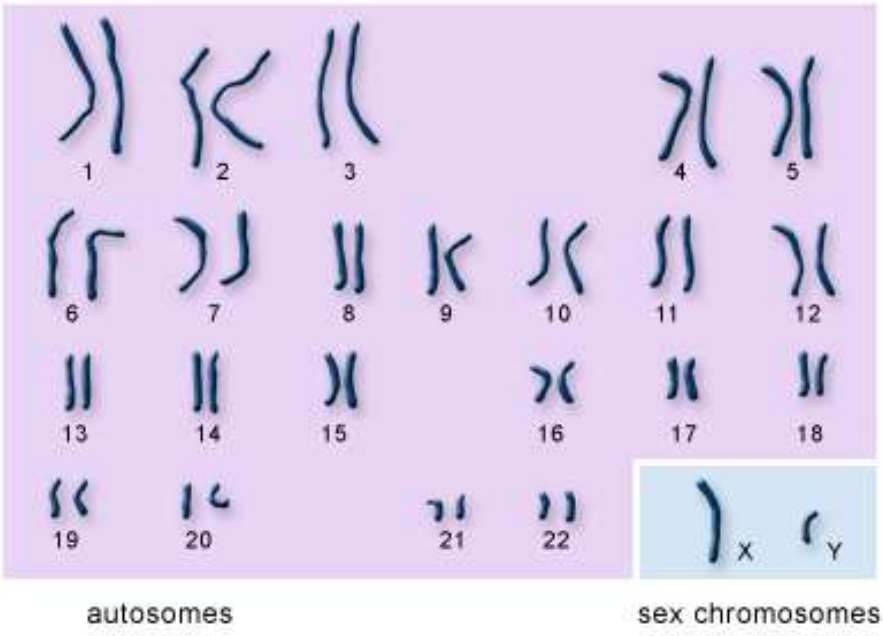
Vemanapuram, Kadapa-516 005, A.P., INDIA
(A State University, Accredited with "A" Grade by NAAC)

DEPARTMENT OF

GENETICS AND GENOMICS

CODE: GENYVUC

M.Sc. Genetics & Genomics Syllabus (NEP-2020)



2025-2026 Onwards

M.SC. GENETICS AND GENOMICS PROGRAM

CURRICULUM

2-Year Master of Science (M.Sc.) with course work in first 3 semesters and project work in 4th semester of II year						
Semester	Courses	Level of Course	Theory		Practical	
			H	Cr	*H*	*Cr*
Semester I	Core Course1: Principles of Genetics	500 -599	4	3	3	1
	Core Course 2: Biomolecules	500 -599	4	3	3	1
	Core Course 3: Cell Biology	500 -599	4	3	3	1
	Skill Oriented Course 1: Analytical Bio-Techniques	400 -499	3	3	3	1
	Skill Oriented Course 2 : Microbial Genetics	400 -499	3	3	3	1
	Open Online TransdisciplinaryCourse1 (Students can choose a 4-week Course from the courses available in the SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/ or Open elective : Basics in Genetics	300 -399		2		
	Indian Knowledge Systems 1 (Either the department can design a 2-credit course or may choose from online https://iksindia.org/courses-offered-by-iks-centers.php) or Indian Perspectives on Health and Lifestyle Management (offered by the Department)	200 -299	2	0	AUDIT COURSE	
	End of Semester I		20	17	15	5
Semester II	Core Course 4: Molecular Genetics	500 -599	4	3	3	1
	Core Course 5: Concepts of Medical Genetics.	500 -599	4	3	3	1
	Core Course 6: Energy Metabolism	500 -599	4	3	3	1
	Skill Oriented Course 3: Genetic Engineering	500 -599	3	3	3	1
	Skill Oriented Course 4: Concepts of Biostatics, Population Genetics.	500 -599	3	3	3	1
	Open OnlineTransdisciplinary Course 2 (Students can choose a 4 weeks Course from the courses available in the ONLINE SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/	300 -399		2		
	Indian Knowledge Systems 2 (Either the department can design a 2-credit course or may choose from online	200 -299	2	0	AUDIT COURSE	

	https://iksindia.org/courses-offered-by-iks-centers.php) Indian Knowledge Systems 2: Bharatiya Knowledge System: Wellness, Health And Conservation Of Natural Resources Used In Indian Medicine(offered by the Department)					
	End of Semester II		20	17	15	5
Semester III	Core Course 7: Epigenetics	500 -599	4	3	3	1
	Core Course 8: Concepts of Genomics	500 -599	4	3	3	1
	Core Course 9: Immunology in Health and Disease	500 -599	4	3	3	1
	Skill Oriented Course 5: Basics of In-silico genomics	500 -599	3	3	3	1
	Skill Oriented Course 6: Pathogenomics	500 -599	3	3	3	1
	End of Semester III		18	15	15	5
Semester IV	Open Online Skill Development Course/s(Students can choose an 8-credit course or 4 Course from the courses available in the ONLINE SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/ https://onlinecourses.swayam2.ac.in/ntr25_ed27/preview	300-399		8		
	Project Work	400-499				12
	End of Semester IV		-	8	-	12
	Total Hours/Credits		58	57	45	27

SEMESTER I

Core Course 1: Principles of Genetics

Course Objectives:
Students will:

1. Master foundational genetic concepts, terminology, and historical milestones
2. Analyze classical and non-Mendelian inheritance patterns
3. Develop experimental skills in genetic analysis and problem-solving

Course Outcomes:
Students will be able to:

1. Explain mechanisms of inheritance and variation
2. Apply genetic principles to real-world scenarios in medicine and agriculture
3. Design and interpret genetic crosses and mapping experiments
4. Diagnose genetic conditions using pedigree analysis and modern tools

Course syllabus

Unit I: Foundations of Genetics

History and key discoveries in genetics. Core terminology and concepts. Applications in medicine, agriculture, and biotechnology. Model organisms: yeast (cell cycle), *Drosophila* (gene mapping), *C. elegans* (development), *Arabidopsis* (plant genetics), mice (disease models). Mendel's laws of inheritance with practical examples. Rediscovery and validation of Mendelian principles.

Unit II: Beyond Mendelism

Non-Mendelian inheritance patterns: incomplete dominance, codominance, multiple alleles. Gene interactions: all types of epistasis and pleiotropy. Variable expressivity and penetrance. Organellar inheritance: chloroplast (leaf variegation) and mitochondrial (*Chlamydomonas*) genetics.

Unit III: Analysis Methods

Classical genetic crosses: test, back, trihybrid and polyhybrid. Human inheritance patterns. Gene-environment interactions. Heterosis mechanisms and applications. Cytogenetics: karyotyping, FISH. Chromosomal variations: polyploidy/aneuploidy.

Unit IV: Transmission Genetics

Sex determination, sex linked genes, sex limited traits, sex influenced traits, Linkage-complete and incomplete linkage, strength of linkage, factors affecting strength of linkage, measurement of linkage strength, importance of linkage; Recombination and crossing over; Chromosome mapping-crossing over as measure of genetic distance, Morgan's cross and strut's event map; recombination mapping with two point test cross, determination of gene order by three point test cross; Mapping of genes by tetrad analysis; Interference and the coefficient of coincidence; Linkage analysis in humans (pedigree analysis). Structural chromosomal changes: inversions, translocations.

Recommended Books:

1. Principles of genetics, by Snustad . Simmons.7thEd.2019.
2. Genetics *A molecular approach* by Peter J.Russell.3ndEd.2016.
3. Genetics fundamentals and applications by Srivastava and Debmalya barch.1stEd.2008.
4. Genetics by Winter, Hickey and Fletcher.2ndEd.2003.
5. Genetics Analysis of genes and genomes by Jones and Bartlett.6thEd.2005.
6. Genetics, Schaum's outlines. 5thEd. TATA McGraw-hilledition.2010.
7. Principles of genetics by Robert H. Tamarin. 7thEd. TATA McGraw-hilledition. 2010.
8. Genetics *A conceptual approach* by Benjamin A. Pierce , 7th Ed.2019.
9. Genetics *Analysis and principles* by Robert J.Brooker 7th Ed.2021.
10. Genomes 4 by T.A.Brown (2017).
11. iGenetics*A molecular approach*, second ed. By Peter J.Russell (2009).

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1. https://nptel.ac.in/courses/121/106/121106008/
2. https://nptel.ac.in/courses/102/104/102104052/
3. https://www.toppr.com/guides/biology/human-reproduction/gametogenesis-in-humans/
4. https://www.ndsu.edu/pubweb/~mcclean/plsc431/mutation/mutation4.htm

Core Course 2: Biomolecules

Course Objectives:

Students will:

1. Master the structural and functional properties of biological macromolecules
2. Analyze metabolic pathways and their regulation
3. Develop experimental skills in biochemical analysis

Course Outcomes:

Students will be able to:

1. Explain structure-function relationships in biomolecules
2. Apply biochemical principles to cellular processes and disease mechanisms
3. Design and interpret biochemical experiments and genetic crosses
4. Utilize key techniques in biomolecular analysis

Course syllabus

UNIT-I: Carbohydrates

Carbohydrates: Introduction of biomolecules, importance of carbohydrates, classification of carbohydrates. Monosaccharides: glucose, galactose and fructose. Glycosidic bond formation. Disaccharides: lactose, maltose and sucrose. Polysaccharides: starch, cellulose, glycogen, peptidoglycans, glycoproteins and mucopolysaccharides.

UNIT-II: Amino Acids and Proteins

Amino acids: Functions, classification, structure and physiochemical properties. Peptide bond formation. Biologically important peptides: Glutathione,Aspartame, Insulin, Glucagon, and Oxytocin. Proteins: biological functions. Structural organization: primary, secondary, tertiary and quaternary structure. Ramachandran plot. Structural and functional relationships. Denaturation and renaturation. Ribonuclease as model protein.

UNIT-III: Lipids and Nucleic Acids

Lipids: Functions of Lipids. Types of Lipids: Triglycerides, waxes, phospholipids,. Lipoproteins. Prostaglandins, leukotrienes, thromboxanes. Steroids and bile acids. Nucleic acids: Purine and pyrimidine bases. Nucleosides and nucleotides. Phosphodiester bond formation and stability. DNA structure: Watson-Crick model. DNA types (A,B, Z). RNA types. DNA denaturation and renaturation. Melting curves.Functions of Nucleic acids.

UNIT-IV: Enzymes

Classification and nomenclature: six main enzyme classes. Enzyme kinetics: Michaelis-Menten equation, Vmax and Km determination. Lineweaver-Burk plots. Factors affecting activity: substrate concentration, pH, temperature. Enzyme inhibition: reversible (competitive, noncompetitive) and irreversible. Allosteric inhibition.

Recommended Books:

1. Principles of Biochemistry by LEHNINGER 8th edition, NELSON &COX (Worth) Publ. 2021.
2. Biochemistry by Jeremy M. Berg, John L. Tymoczko, Gregory J, Gatto. 9th edition (2023) Publ: W H Freeman
3. Principles of Biochemistry by Donald Voet and Judith G. Voet .– 5th Ed.. John Willey and Sons,Inc. 2018 .
4. Biochemistry by Jeremy M Berg, Lubert Stryer, John Tymoczko, Gregory Gatto W.H. Freeman andcompany,9th Ed. NewYork, 2022.
5. Harper's Illustrated Biochemistry Thirty-second Edition (A & L LANGE SERIES),2021.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1 Biochemistry of Biomolecules.
https://onlinecourses.swayam2.ac.in/cec20_bt12/preview
2 Biomolecules: Structure, Function in Health and Disease.
http://ugcmoocs.inflibnet.ac.in/ugcmoocs/view_module_ug.php/353
3 Introduction to Carbohydrates.
https://aklectures.com/lecture/carbohydrates/introduction-to-carbohydrates

Core Course 3: Cell Biology

Course Objectives:

1. To analyze the structural and functional organization of prokaryotic and eukaryotic cellular components
2. To examine membrane dynamics, transport mechanisms, and cell signaling pathways
3. To investigate cell cycle regulation and division processes in eukaryotes

Course Outcomes:

By completing this course, students will be able to:

1. Distinguish prokaryotic and eukaryotic cellular architectures with emphasis on organelle ultrastructure and functions
2. Evaluate biomembrane properties using the fluid mosaic model and analyze transport systems (Na+/K+ ATPase, Ca2+ ATPase)
3. Map cell signaling pathways involving second messengers (cAMP, Ca2+, NO) and their regulatory mechanisms
4. Demonstrate the molecular control of cell cycle phases (including checkpoints and kinase regulation)

Course syllabus

UNIT-I: Cell Structure and Organelles

Structural organization of prokaryotic and eukaryotic cells. Ultrastructure and functions of cellular components: nucleus, mitochondria, plastids, endoplasmic reticulum, Golgi complex, lysosomes, microbodies, and ribosomes. Cytoskeletal elements: microtubules and microfilaments and their roles in cell structure and motility.

UNIT-II: Cellular Diversity and Membranes

Comparative analysis of plant and animal cell structure and function variations. Tissue types with focus on epithelial tissues and their basement membrane. Extracellular matrix components: collagen, elastin, and fibrillin. Nuclear structures: chromatin organization, telomeres, and centromeres. Membrane dynamics: cell receptors, endocytosis, and exocytosis processes. Biomembrane composition: detailed study of plasma and organelle membranes including their lipid, protein, and carbohydrate constituents. Membrane architecture: molecular structure, fluidity characteristics, and the fluid mosaic model.

UNIT-III: Membrane Transport and Cell Signaling

Mechanisms of membrane transport with emphasis on active transport systems: Na+/K+ ATPase and Ca2+ ATPase pumps. Fundamentals of cell signaling: signaling molecules and pathways. Second messenger systems: calcium/calmodulin, inositol phosphate, cAMP, cGMP, and NO signaling. Signal transduction cascades and their regulation, including inhibition mechanisms of signaling pathways.

UNIT-IV: Cell Cycle Regulation

Eukaryotic cell cycle phases and their control mechanisms: protein kinase roles, checkpoint regulation, kinase inhibitors, and cellular responses. Detailed examination of M-phase events: prophase (chromosome condensation, spindle formation, nuclear envelope breakdown), prometaphase, metaphase (chromosome alignment, microtubule dynamics), anaphase (proteolytic regulation, chromosome separation), and telophase (nuclear reformation). Cytokinesis processes and forces involved in cell division. Meiosis stages and regulatory mechanisms, with special focus on oocyte meiosis and fertilization processes.

Reference Books:

1. Karp's Cell Biology, 8th Edition, Global Edition.Gerald Karp, Janet Iwasa, Wallace Marshall:
2018.
2. The Cell a molecular approach by G M Cooper &E Hausman, 6thEd. Sinauer Associates Inc.,U.S.; 7th edition,2015.
3. Molecular Cell Biology by Lodish et al., 2008, 9th Ed. W.H Freeman andCompany,2021.
4. Essential Cell Biology by Alberts et al, 4th Ed. Garland Science, Taylor & Francis group, 2014.

Related Online Contents [
MOOC, SWAYAM, NPTEL, Websites etc
1.https://bio.libretexts.org/Bookshelves/Cell_and_Molecular_Biology/Book%3A_Cells_-Molecules_and_Mechanisms_(Wong)
2 https://nptel.ac.in/courses/102/106/102106025/#
3 https://www.easybiologyclass.com/molecular-biology-online-tutorials-lecture-notes-studymaterials/

Skill Oriented Course 1: Analytical Bio-Techniques

Course Objectives:

1. To impart fundamental knowledge of pH measurement, buffer systems, and centrifugation techniques in biological research
2. To develop expertise in separation techniques including chromatography and electrophoresis
3. To understand principles of radioactivity and its applications in biological systems
4. To master microscopy and spectroscopy techniques for biomolecular analysis

Course Outcomes:

Upon completion, students will be able to:

1. Operate pH meters and electrodes, prepare buffer solutions, and apply centrifugation techniques appropriately.
2. Select and perform suitable chromatographic and electrophoretic methods for biomolecule separation and purification.
3. Utilize various microscopy and spectroscopy techniques for biological sample analysis.
4. Apply Beer-Lambert law in spectroscopic quantification of biomolecules.

Course syllabus

UNIT-I: pH, Buffers and Centrifugation

pH and Buffers: pH meter, glass electrode, reference electrodes, ion selective electrodes and oxygen electrode. Measurement of pH, pKa. Importance of pH and buffers in biological systems, Solutions - Molarity, Molality, Normality.

Centrifugation: Basic principles of centrifugation, Preparative and analytical, Rotors: Fixed Angle and Swinging Bucket, Differential Centrifugation, Ultra-centrifugation. Centrifuge- uses and Safety.

UNIT-II: Chromatography and Electrophoresis

Chromatography: Principles, method and applications of Chromatography: Paper Chromatography, Thin Layer Chromatography (TLC), Column Chromatography, Gas liquid chromatography (GLC), High Performance Liquid Chromatography (HPLC), Affinity Chromatography.

Electrophoresis: Principles, method and applications of Agarose Gel Electrophoresis, Sodium Dodecyl Sulphate Polyacrylamide Gel Electrophoresis (SDS-PAGE), Native PAGE, Immuno-electrophoresis, Isoelectric - Focusing and Two-Dimensional Gel Electrophoresis; Blotting techniques: Northern blotting, Southern blotting, Western blotting.

UNIT-III: Microscopy and Radioisotope Techniques

Microscopy: Principles, method and applications of light microscopy, Phase contrast Microscopy, Transmission electron microscopy (TEM), Scanning electron microscopy (SEM) and fluorescence microscopy.

Radioactivity: Half –life, decay constant, average life, units of radioactivity, Isotope dilution techniques and Radioactive disposal. Biological effects of radiation, applications of Radioisotopes in medicine.

UNIT-IV: Spectroscopy

Spectroscopy: Electromagnetic radiations, Beer – Lamberts law, principles and applications of colorimetry, spectrophotometry: UV, fluorimetry, flame photometry, Infrared, ESR, NMR spectroscopy, Mass spectroscopy and XRD.

Recommended Books:

1. Principles and Techniques of Practical Biochemistry, Ed. Williams andWilson.,8th edition-April-2018.
2. Biophysical chemistry, principle and Techniques,Himalaya Pub. House, 2009.
3. An Introduction to Practical Biochemistry. David t. Plummer, Tata Mac grew – Hill 3rd Ed.2017.
4. Physical Biochemistry: Applications to Biochemistry and Molecular Biology (Life Sciences/Biochemistry), 1983.
5.Analytical Techniques in Biochemistry and Molecular Biology by Rajan Katoch, Springer,2011.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1 http://epgp.inflibnet.ac.in/
2 http://epgp.inflibnet.ac.in/Home/ViewSubject?catid=944

Skill Oriented Course 2: Microbial Genetics

Course Objectives:
Students will:

1. Master fundamental concepts of bacterial and viral genetics
2. Analyze mechanisms of horizontal gene transfer and viral replication
3. Develop skills in microbial genetic analysis and interpretation

Course Outcomes:
Students will be able to:

1. Explain bacterial genetic systems and their applications
2. Compare different mechanisms of gene transfer in prokaryotes
3. Classify viruses and describe their replication strategies
4. Design experiments to study microbial genetics
5. Evaluate emerging concepts in viral evolution and subviral agents

Course syllabus

UNIT-I: Bacterial Genetics Fundamentals

Essentials of bacterial genetics: General properties of bacteria. Growth and culture of bacteria: nutritional requirements, pure cultures and methods of isolation of pure cultures, synchronous cultures, continuous cultures. Genetic nomenclature. Mutants and mutations: Types of mutants, isolation and characterizations of mutants, revertant & reversion, uses of mutations;

UNIT-II: Bacterial Gene Transfer Mechanisms

Transformation: Transformation in the history of molecular biology, uptake of DNA in transformation and Integration of transforming DNA. Transduction: Generalized transduction and specialized transduction. Conjugation: Essential features and mechanism of DNA transfer during conjugation, Formation of Hfr strains, Use of Hfr strains in genetic cross, transfer of chromosomal genes to F plasmid.

UNIT-III: Viral Classification and Replication

Classification and replications of viruses: General properties, nucleic acid-based classification (Baltimore classification), Replication. DNA viruses: SV40, CaMV, Bacteriophage T4 and λ. RNA viruses: TMV, Influenza. Retroviruses HIV hepatitis B virus.

UNIT-IV: Basics of Fungal Genetics

Fungi: Nutrition and growth physiologies & Structure, Fungal Morphology and spores, Fungal Nuclei and chromosomes, Haploidy and Nuclear stages. Cell cycle, Homo and hetero karyons reproduction in fungi; Conidia, sexual reproduction plasmogamy, Karyogamy, Meiosis, Para sexual cycle, Eukaryotic genome: *Soucharomyes cerevisiae*, Wild types and mutants, Maushroom life cycle.

Recommended Books:

1. Microbilal genetics by Stanley R.Maloy, John E.Cronan and David Frieflder(1994), second ed.,Jones
and Bartlettpublishers.
2. Molecular genetics of bacterial by Larry syder and Wendy champnes (2013), 4th Ed., ASM press.
3. Modern virology by N.J.Dimmock, A.J.Easton and K.N.Leppard (2016), 7thEd., Blackwell publishing.
4. Principles of Virology, byS.J.Flint et al., 5th Ed.ASM press, 2020.
5. Principles of Molecular Virology by A.Cann , 6th Ed. Academic Press,2015.
6. Brok Biology of Microorganisma. 2009 pearson Madigan Martinko Dunlap clark
7.Essential Fungal Genetics. Davis Moore Lilyann Novak Frazer SPRINGER. ISBN:0-387-95367-1

Open Online Transdisciplinary Course1

(Students can choose a 4-week Course from the courses available in the SWAYAM&SWAYAM PLUS)https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/

Open Elective: Basics in Genetics

Course Objectives:

1. To introduce the fundamental biomolecules (carbohydrates, lipids, proteins, nucleic acids) and their biological significance.
2. To compare and contrast the structural and functional differences between bacterial, plant, and animal cells.
3. To explain the structure and function of genetic material (DNA, RNA) and chromosome organization.
4. To describe the processes of mitosis and meiosis and their importance in growth and heredity.

Course Outcomes:

Upon completion of this course, students will be able to:

1. Identify and describe the roles of key biomolecules in cellular processes.
2. Distinguish between prokaryotic and eukaryotic cell structures and their functional components.
3. Explain the structural features of chromosomes and their role in genetic inheritance.
4. Summarize the stages and significance of mitosis and meiosis in cell division.
5. Apply Mendelian principles to predict inheritance patterns and solve basic genetic problems.

UNIT-I: Biomolecules and Genetic Basis of Life

Overview of Biomolecules: Fundamental classes of biomolecules and their significance in sustaining life processes – carbohydrates, lipids, proteins, and nucleic acids. Structure, types, and biological functions of DNA and RNA; roles of nucleic acids as carriers of genetic information. Introduction to enzymes and hormones in regulating metabolic activities.

UNIT-II: Cellular Architecture and Organization

Prokaryotic Cell Structure – Basic organization of prokaryotic cells with focus on bacterial cell wall, cell membrane, and arrangement of DNA. **Eukaryotic Cells** – Structure of plant and animal cells; functions of major organelles such as endoplasmic reticulum, Golgi apparatus, mitochondria, plastids, and nucleus.

UNIT-III: Chromosomes and Cell Division

Structure and Organization of Eukaryotic Chromosomes, Chromatin Types: Euchromatin and Heterochromatin,Key Chromosomal Features: Telomeres, Centromeres, Homologous Pairs, Cell Cycle: Mitosis (Stages and Significance) and Meiosis (Haploid vs. Diploid Cells), Stem Cells and Their Biological Importance

UNIT-IV: Principles of Heredity and Genetic Inheritance

Historical Milestones in Genetics, Mendelian Genetics: Terminology and Experimental Foundations, Laws of Dominance, Segregation, and Independent Assortment, Monohybrid Crosses and Inheritance Patterns, Overview of Non-Mendelian Inheritance Mechanisms.

Recommended Books:

1. Snustad, D.P., & Simmons, M.J. *Principles of Genetics*. 7th Ed., 2015.
2. Atherly, A.G., Girton, J.R., & McDonald, J.F. *The Science of Genetics*. 1999.
3. Russell, P.J. *Genetics: A Molecular Approach*. 3rd Ed., 2016.
4. Srivastava, P.K., & Barch, D. *Genetics: Fundamentals and Applications*. 1st Ed., 2008.
5. Winter, P.C., Hickey, G.I., & Fletcher, H.L. *Genetics*. 2nd Ed., 2003.
6. Hartl, D.L., & Ruvolo, M. *Genetic Analysis of Genes and Genomes*. 6th Ed., Jones and Bartlett, 2005.
7. Stansfield, W.D., & Coliver, J. *Schaum's Outline of Genetics*. 5th Ed., Tata McGraw-Hill, 2010.
8. Tamarin, R.H. *Principles of Genetics*. 7th Ed., Tata McGraw-Hill, 2010.

IndianKnowledgeSystems1

(Either the department can design a 2-credit course or may choose from onlinehttps://iksindia.org/courses-offered-by-iks-centers.php)

IKS 1: Indian Perspectives on Health and Lifestyle Management

Course Objectives:

1. To enable students to critically analyze and appreciate diverse Indian health philosophies and their application in personal and societal lifestyle management.
2. To develop independent thinking in proposing improvements in food habits, cultural practices, psychological well-being, environmental harmony, and economic sustainability.
3. To equip students with practical knowledge to assess physiological and clinical health status and suggest appropriate diet and nutritional therapies based on Ayurvedic research and experiential wisdom.
4. To foster interpersonal sensitivity and an understanding of behavioral change mechanisms to build healthy relationships and promote holistic well-being.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. Understand foundational concepts of Ayurveda and its role in preventive health care.
2. Differentiate between **daily (Dinacharya)** and **seasonal (Ṛtucaryā)** lifestyle practices.
3. Analyze and interpret traditional health knowledge and integrate it with contemporary lifestyle needs.
4. Plan and manage nutritional and lifestyle interventions for different demographic groups.
5. Recognize the professional scope and relevance of continuously upgrading knowledge in traditional health sciences.

Course Syllabus

1. Course Overview and Structure
2. The Concept and Importance of Health
3. Introduction to Ayurveda
4. Philosophical Foundations of Ayurveda
5. Classical Definition of Health
6. The Tridosha Theory (Vata, Pitta, Kapha)
7. The Seven Dhatus (Tissues) and Their Functions
8. Introduction to Yoga and its Integrative Role
9. Dinacharya – The Ayurvedic Daily Regimen
10. Ṛtucaryā – Seasonal Lifestyle Practices
11. The Science of Habits and Behavioral Conditioning
12. Principles of Food Intake (Ahara Vidhi) – Part I
13. Principles of Food Intake – Part II
14. Introduction to Siddha Medicine
15. Introduction to Homeopathy
16. The Science of Habits – Continuation & Case Applications
17. Common Ayurvedic Home Remedies for Preventive Care

Recommended Books:

1. Valiathan, M.S. (2015). *The Legacy of Caraka*. University Press (India), Hyderabad. ISBN: 978-81-7371-667-6
2. Valiathan, M.S. (2015). *The Legacy of Susruta*. University Press (India), Hyderabad.
3. Valiathan, M.S. (2015). *The Legacy of Vagbhata*. University Press (India), Hyderabad.
4. Wujastyk, D. (2001). *The Roots of Ayurveda*. Penguin Classics, Haryana, India. ISBN: 978-0140436808
5. Jayesh Thakker, S. Chaudhari, Prasanta S. Sarkar (2011). *Ritucharya: Answers to the Lifestyle Disorders*, AYU - An International Quarterly Journal of Research in Ayurveda, 32(4), 466–471. DOI: 10.4103/0974-8520.96117

Core Practical-I (Core 1+2+3)

1.Principles of Genetics

i. Problems in Genetics.
ii. Microscopic observation of chromosomes.
iii. Karyotyping.
iv. Development of point mutations
v. Development of deletion mutations

2. Biomolecules

i. Qualitative tests for identification of Carbohydrates.
ii. Qualitative tests for identification of amino acids
iii. Qualitative tests for identification of nucleic acids.
iv. Qualitative tests for identification of Lipids.
v. Quantitative tests for protein estimation (Lowry / Bardfoed's methods).
vi. Quantitative tests for glucose estimation (DNS /Nelson Somagy Method).
vii. Quantitative tests for Free Amino acids estimation (Ninhydrin Method).
viii. Quantitative tests for Cholesterol estimation (Zak's method).
ix. Quantative estimation of Bilirubin by Vanden Bergh method.
x. Glycine estimation by Sorensen method.
xi. Estimation of DNA by diphenylamine method.
xii. Estimation of RNA by orcinol method.

3. Cell Biology

i. Preparation of plant and animal cells slides for observation under microscope.
ii. Preparation of blood smear and differential staining of blood cells.
iii. Study of divisional stages in Mitosis.
iv. Study of divisional stages in Meiosis
v. Estimation of chlorophyll in the leaf extract.
vi. Isolation of chloroplasts by sucrose density gradient centrifugation

Skill Practical I (Skill 1+2)

1. Analytical Bio-Techniques

i. Preparation of buffers and measurement of pH.
ii. Verification of Beer's Law and determination of λ max for color solutions.
iii. Paper chromatography for separation of amino acids and plant pigments.
iv. Thin layer chromatography (TLC) for separation of lipids and amino acids.
v. Protein precipitation by Ammonium Sulphate and Dialysis.
vi. SDS-PAGE for separation of proteins.
vii. Submarine Agarose gel electrophoresis for DNA separation.
viii. Ion-exchange column chromatography (Demonstration).
ix. Gel permeation column chromatography (Demonstration).

2. Microbial Genetics

i. Bacterial growth curve.
ii. Isolation of mutants by gradient plate technique.
iii. Isolation of mutants by replica plate technique.
iv. Bacterial survival curve in the presence of UV.
v. Isolation of mutations in bacteria by physical agents.
vi. Isolation of mutations in bacteria by chemical agents.
vii. Bacterial conjugation and transformation.

SEMESTER II

Core Course 4: Molecular Genetics

Course Objectives:

1. To establish fundamental understanding of genetic material and nucleic acid biochemistry
2. To analyze genome organization and DNA replication mechanisms across organisms
3. To examine molecular processes of transcription, translation, and their regulation
4. To investigate DNA recombination, repair, and gene expression control systems

Course Outcomes:
Upon completion, students will be able to:

1. Demonstrate comprehensive knowledge of nucleic acid chemistry and topology
2. Compare genome organization and replication strategies in prokaryotes and eukaryotes
3. Explain transcriptional and translational mechanisms with their regulatory controls
4. Analyze recombination processes and DNA repair pathways
5. Apply operon concepts to predict gene expression patterns

Course syllabus

UNIT-I: Genetic Material and Nucleic Acids

Genetic material: Functions of the genetic material; Evidence for DNA is the genetic material; Evidence for RNA is the genetic material. Nucleic acids: Types, chemistry and structure of DNA and RNA; Chargaff's ratios; DNA topology-linking, writhing, twisting number, positive and negative supercoiling, significance of supercoiling in vivo, topoisomerases types and mechanism; Unusual DNA secondary structures.

UNIT-II: Genome Organization and Replication

Genome organization: Bacterial genome, Eukaryotic Chromosome-Histone proteins, non-histone chromosomal proteins, nucleosome, linkers, chromatin fibers, centromeres, telomeres and alternative chromatin structures. DNA replication: Concepts and strategies/models for replication. Relation between cell cycle and DNA replication. Molecular mechanisms of DNA replication in prokaryotes and eukaryotes. DNA damage andrepair.

UNIT-III: Recombination and Transcription

Recombination: Homologous recombination, site-specific recombination and transposition. Gene expression (Transcription and translation): Overview of gene expression. Transcription (RNA biosynthesis): Types of RNA and their role; RNA polymerases of prokaryotes and eukaryotes. Transcription mechanism in pro and eukaryotic cells-promoter recognition, initiation, elongation and termination of RNA synthesis. Maturation and processing of different RNA transcripts-capping, methylation, polyadenylation, splicing, RNA editing and modification of nucleosides in tRNAs.

UNIT-IV: Translation and Gene Regulation

Translation (protein biosynthesis) : Genetic code and its elucidation, structure and composition of prokaryotic and eukaryotic ribosome's; Structural features of rRNA, mRNA and tRNAs in relation to function, mechanisms of protein biosynthesis in prokaryotic and eukaryotes; post- translational modification of proteins and their sorting and targeting and degradation; regulation of translation; inhibitors of protein biosyntheses.

Regulation of gene expression: Principles of gene regulation, terminology and operon concepts, enzyme induction and repression; positive and negative regulation in E.coli–lac, and regulation by attenuation-trp operons.

Recommended Books:

1) Fundamental Molecular Biology by Lizabeth A.Allison. 3rd Ed.Blackwell publishing, 2021.
2) Molecular Biology of the gene by Watson et al.,7th Ed, ,Addison WesleyLongman. 2022
3) Principles of Biochemistry by LEHNINGER 8th edition, NELSON &COX (Worth) Publ. 2021.
5) Instant notes in Molecular Biology by P.C.Turner et al.Viva BooksPvt.Ltd. 2016.
6) Advanced Molecular Biology by A Concise reference. by R.M.Twyman. Viva Book Pvt.Ltd. 1998.
7)Molecular Biology of the Gene by Curtis Holmes,2019.
8) Molecular Biology by David Fdreifelder,1995 NarosaPubl.House.
9) Molecular Cell Biology by Lodish et al. 9th ed, Scientific American books, W.H.Freeman and Company 2021.
10) Genes VIII by Lewin,Oxford University Press. 2017.

Core Course 5: Concepts of Medical Genetics

Course Objectives:

1. To analyze the molecular and chromosomal basis of genetic disorders
2. To classify genetic diseases by their inheritance patterns and clinical manifestations
3. To evaluate modern genetic testing methodologies and their clinical applications
4. To develop management strategies for genetic disorders incorporating genetic counselling

Course Outcomes

After completing this course, students should be able to:

1. Explain the causes of different genetic disorders (chromosomal and gene mutations) with examples.
2. Describe how complex disorders (like diabetes) and organ-specific disorders are inherited.
3. Choose the right genetic tests (like karyotyping) for diagnosing disorders.
4. Plan treatment approaches including diet changes, enzyme therapies, and gene therapies.
5. Use genetic counselling to help families understand and manage genetic conditions.

Course syllabus

UNIT-I: Reasons for Genetic Disorders

Occurrence and classification of Genetic disorders. Chromosomal mutations: Structural changes - duplications, deletions, inversions, and translocations. Numerical abnormalities:Aneuploidy and polyploidy. Gene mutations: Base substitutions, insertions, and deletions. Transposable elements: SINEs and LINEs and genetic variation. Monogenic disorders (cystic fibrosis, Huntington's disease, hemophilia), chromosomal disorders (Down syndrome, cri-du-chat syndrome), and inborn errors of metabolism (alkaptonuria, cystinuria).

UNIT-II: Complex and Organ-Specific Disorders

Polygenic disorders: diabetes mellitus and atherosclerosis. Mitochondrial disorders: maternal inheritance patterns. DNA repair defects: xeroderma pigmentosum. Organ-specific genetic disorders: eyes (color blindness, retinitis pigmentosa), skeletal system, and skin. Genomic imprinting disorders: Prader-Willi syndrome, Angelman syndrome, and Beckwith-Wiedemann syndrome.

UNIT-III: Genetic Testing Methods

Prenatal diagnosis: ultrasonography and maternal serum screening. Invasive procedures: amniocentesis and chorionic villus sampling. Genetic testing methods: protein truncation tests, SSCP analysis, DNA sequencing, karyotyping, and chromosomal microarrays.

UNIT-IV: Management and Genetic Counseling

Dietary management: metabolic disorders. Enzyme replacement therapies: specific enzyme deficiencies. Emerging treatments: gene therapy applications. Symptomatic management: clinical manifestations. Genetic counseling: risk assessment, family planning guidance, and psychosocial support. Multidisciplinary care teams: optimize patient outcomes.

Recommended Books:

1. Human Genetics by Gardner and Davies, 2nd Ed. 2012
2. Elements of Medical Genetics, Emery's. 15th Ed. 2017
3. Medical genetics / Lynn B. Jorde, John C. Carey, Michael J. Bamshad. 6th Ed. 2020
4. Human Genetics by Gangane, 6th Ed. 2021.
5. Principles of Genetics, by Snustad. Simmons. 7th Ed. 2015.

Core course 6: Energy Metabolism

Course Objectives

By the end of this course, students will:

1. Understand thermodynamic principles (Gibbs free energy, redox reactions) and their role in cellular energy production.
2. Learn the pathways of carbohydrate metabolism (glycolysis, TCA cycle, gluconeogenesis) and their regulation.
3. Explore specialized carbohydrate pathways (glycogen metabolism, pentose phosphate pathway) and related metabolic disorders.
4. Examine lipid metabolism, including fatty acid synthesis/degradation, cholesterol metabolism, and ketone body formation.

Course Outcomes

After completing this course, students should be able to:

1. Explain bioenergetic principles (ΔG, ATP synthesis) and mitochondrial energy production (ETC, oxidative phosphorylation).
2. Describe carbohydrate metabolic pathways (glycolysis, TCA cycle) and their regulatory mechanisms.
3. Analyze specialized pathways (glycogen metabolism, PPP) and disorders (e.g., galactosemia, glycogen storage diseases).
4. Illustrate lipid metabolism (fatty acid oxidation, cholesterol synthesis) and its connection to energy homeostasis.
5. Correlate metabolic dysregulation (e.g., diabetes, hyperlipidemia) with pathway defects.

Course syllabus

UNIT-I: Fundamentals of Bioenergetics & Cellular Energy Production

Bioenergetics: Thermodynamic principles – Chemical equilibria; Gibbs free energy(G), enthalpy (H), entropy (S). Free energy change in biological transformations in living systems; Actual free energy change, standard free energy change. High energy compounds. Mitochondrial electron transport system-Electron carriers in the respiratory chain, organization of Enzyme complexes and their importance. Substrate level phosphorylation and oxidative phosphorylation. Respiratory chain inhibitors and uncouplers of oxidative phosphorylation. ATP synthesis and regulation of ATP producing pathways. ATP yield from complete oxidation of Glucose. Regulation of oxidative phosphorylation.

UNIT-II: Carbohydrate Metabolism

Metabolism of carbohydrates: Glycolysis: Preparative and payoff phases of Glycolysis, Regulation of glycolysis, Fermentation: the anaerobic fate of pyruvate, Metabolism of hexoses other than glucose: fructose, galactose and mannose, Citric acid cycle: pyruvate dehydrogenase complex, metabolic sources of acetyl CoA, reactions and regulation of citric acid cycle. Amphibolic nature of citric acid cycle.

UNIT-III: Specialized Carbohydrate Pathways & Metabolic Disorders

Other pathways of carbohydrate metabolism: Uronic acid pathway, metabolism of amino sugars, glycogen metabolism: Glycogen synthesis and break down, Regulation of glycogen synthesis and breakdown. Gluconeogenesis and maintenance of blood glucose levels, glyoxylate cycle. Pentose phosphate pathway of glucose oxidation, Disorders of carbohydrate metabolism – Glycogen, galactose, Fructose.

UNIT-IV: Lipid Metabolism

Lipid Metabolism: Biosynthesis and degradation of fatty acids, palmitic acid.Energy yield and regulation of palmitic acid, Biosynthesis of triacyl glycerols, and membrane phospholipids. Biosynthesis and degradation of cholesterol and its regulation. Metabolism of lipoproteins and Ketone bodies.

Recommended Books:

1. Principles of Biochemistry by LEHNINGER 8th edition, NELSON &COX (Worth) Publ. 2021.
2. Principles of Biochemistry by Donald Voet and Judith G. Voet .– 5th Ed.. John Willey and Sons,Inc. 2018 .
3. Biochemistry by Jeremy M Berg, Lubert Stryer, John Tymoczko, Gregory Gatto W.H. Freeman andcompany,9th Ed. NewYork, 2019.
4. Harper's Illustrated Biochemistry Thirty-First Edition (A & L LANGE SERIES),2018.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1.https://www.researchgate.net/publication/322407168_Machine_Learning_Methods_for_Analysis_of_Metabolic_Data_and_Metabolic_Pathway_Modeling
2 https://www.nature.com/articles/s41540-018-0054-3
3https://www.researchgate.net/publication/263474674_A_Comprehensive_View_on_Metabolic_Pathway_Analysis_Methodologies
4 https://www.classcentral.com/course/swayam-enzymology-19860
5 https://www.udemy.com/course/enzymology/

Skill Oriented Course 3: Genetic Engineering

Course Objectives

By the end of this course, students will:

1. Understand the tools and techniques used in genetic engineering (restriction enzymes, vectors, PCR).
2. Learn gene cloning strategies and DNA analysis methods (sequencing, mutagenesis).
3. Gain knowledge hands-on experience in recombinant DNA technology and protein expression.

Course Outcomes

After completing this course, students should be able to:

1. Explain how genetic engineering tools work and their applications.
2. Perform basic genetic engineering techniques (PCR, gel electrophoresis, bacterial transformation).
3. Design and execute a gene cloning experiment.
4. Analyze DNA and protein data from genetic engineering experiments.
5. Apply genetic engineering concepts to real-world problems (vaccines, gene therapy, biotechnology).

Course syllabus

UNIT-I: Tools for Genetic Engineering

Tools for genetic engineering: Introduction to Genetic engineering; Enzymes - Restriction nucleases (exo- and endonuclease), Restriction endonuclease: Nomenclature, classification, cleavage pattern and applications; Enzymes in modification- Polynucleotide phosphorylase, DNase (DnaseI, DNaseII, Exonuclease III and Mung bean nuclease), Phosphatases, Methylases, Ligases, Polynucleotide kinase and RNase. Oligonucleotides- primers, linkers and adaptors; Vectors for cloning- types, plasmid and phage vectors, cosmids, phagemids, BAC & YAC.

UNIT-II: PCR for Gene Amplification and Detection

PCR for gene amplification and detection: PCR principle and mechanism, Enzymes of PCR-Taq polymerase, Reverse Transcriptase, factors affecting PCR, different types of PCR (RT-PCR, nested PCR, Multiplex PCR and real time PCR) and their applications, Probes: Oligonucleotide, DNA and RNA probes, methods for radioactive and non-radioactive labeling; Strategies for molecular cloning: Choice of vector for cloning, preparation of DNA molecules for cloning, ligation, transformation into bacterial cells, screening and identification of positive clones.

UNIT-III: DNA Libraries and Sequencing

Libraries: Construction and screening of cDNA and genomic DNA libraries; DNA sequencing- Chemical method of Maxam and Gilbert, Sanger's dideoxy chain termination and automated sequencing; Site-directed mutagenesis: Oligonucleotide directed mutagenesis, site-directed mutagenesis by means of the PCR and importance of site-directed mutagenesis.

UNIT-IV: Gene Expression and Applications

Gene expression: Construction of vectors for expression- choice of promoter, ribosome binding sites, transcription terminator, fusion protein tags, purification tags, protease cleavage sites and reporter genes; Over expression of heterologous protein in bacterial, purification and detection and analysis of recombinant protein. Vaccines-Types of vaccines, subunit vaccines, peptide vaccine, vector vaccines. Gene therapy- Ex vivo and In vivo gene therapy methods; Applications of genetic engineering.

Reference Books:

1. Principles of Gene Manipulation and genomics: An Introduction to genetic engineering by Primrose and Twyman, 7th Ed. . 2013.
2. Fundamental Molecular Biology by Lizabeth A.Allison. 3rd Ed.Blackwell publishing, 2021.
3. Molecular Biotechnology: Principles and Applications of Recombinant DNA (ASM Books) by Bernard R. Glick, Cheryl L. Patten 6th Ed.2022.
4. Advanced Molecular Biology by A Concise reference.1998, by R.M.Twyman. Viva BookPvt.Ltd.
5. Fundamental Molecular Biology by Lizabeth A.Allison. 3rd Ed.Blackwell publishing, 2021.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1 https://onlinecourses.nptel.ac.in/noc19_bt15/preview
2 https://www.classcentral.com/course/swayam-genetic-engineering-theory-and-application-14090
3 http://biology.kenyon.edu/courses/biol114/Chap08/Chapter_08a.html
Course Designed By: Dr. S. Selvakumar

Skill Oriented Course 4: Concepts of Biostatics and Population Genetics

Course Objectives

By the end of this course, students will:

1. Understand fundamental statistical methods for biological data analysis
2. Learn principles of hypothesis testing and experimental design
3. Examine population genetics concepts
4. Study quantitative genetic inheritance and trait analysis

Course Outcomes

After completing this course, students should be able to:

Statistical Methods

1. Organize and summarize biological data using appropriate statistical measures
2. Apply probability distributions to analyze biological datasets
3. Perform and interpret common statistical tests (t-tests, chi-square, ANOVA)
4. Evaluate relationships between variables using correlation and regression

Population Genetics

5. Calculate allele and genotype frequencies in populations
6. Apply Hardy-Weinberg principles to analyze genetic variation
7. Explain how evolutionary forces alter population genetics

Quantitative Genetics

8. Differentiate between qualitative and quantitative traits
9. Estimate heritability and analyze polygenic inheritance
10. Interpret QTL mapping data

Course syllabus

UNIT-I: Statistical Concepts

Population, sample, variables, classification and Tabulation of date, Diagrams & graphs, frequency distribution, skewness, kurtosis, central tendency, Average, mean, median, mode, Dispersion, Measures of dispersion, Standard deviation, coefficient of Poisson, Normal distribution, standard error.

UNIT-II: Statistical Tests

Hypothesis testing, Null hypothesis, Type-I & Type-II errors, level of significance, Decision about Null hypothesis (HO), Student's 't' test-applications, chi-square fest, Application Analysis of Variance (ANOVA) - F test- Applications Correlation, Types-Applications, Regression- Applications.

UNIT-III: Population Genetics

Population genetics: Fundamental concepts, genetic structure of populations- Genotype frequencies, Allele frequencies; The Hardy-Weinberg Law- Assumption, prediction, deviation and extensions; Genetic variation- in space and time, in natural populations; Forces that changes gene frequencies in population- Mutation, genetic drift, migration; Hardy-Weinberg and natural selection.

UNIT-IV: Quantitative Genetics

Quantitative genetics: Quantitative characteristics-variability, relation between genotype and phenotype, types of quantitative characteristics, phenotypic inheritance, determining gene number for a polygenic characteristic, QTL data analysis; Heritability- phenotypic variance, Types of heritability, calculating heritability, limitations of heritability; Locating genes that affect quantitative characteristics.

Recommended Books:

1.Genetics A conceptual approach by Benjamin A. Pierce , 7th Ed.2019.
2.Genetics Analysis and principles by Robert J.Brooker 7th Ed.2021.
3.Genomes 4 by T.A.Brown (2017).
4.iGenetics A molecular approach, second ed. By Peter J.Russell (2009).
5.Biostatistics: A Foundation for Analysis in the Health Sciences, 11th Edition, Wayne W.Daniel, Chad L. Cross, Wiley, 2018.
6.Biostatistical analysis - Jerrold H.Zar. Pearson Education, 5th Edition Pearson Education,2010.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1.Mathematical Biostatistics Boot Camp 1
https://www.coursera.org/learn/biostatistics
2. Mathematical Biostatistics Boot Camp 1
https://www.coursera.org/learn/biostatistics-2
3. Application of big data
https://journalofbigdata.springeropen.com/articles/10.1186/s40537-019-02

Open Online Trans-disciplinary Course 2

(Students can choose a 4-week Course from the courses available in the SWAYAM&SWAYAM PLUS)https://swayam.gov.in/
https://swayam-plus.swayam2.ac.in/

Indian Knowledge Systems2

(Either the department can design a 2-credit course or may choose from online
https://iksindia.org/courses-offered-by-iks-centers.php)

IKS 2: Bharatiya Knowledge System: Wellness, Health And Conservation Of Natural Resources Used In Indian Medicine

Focus Area/Subject Area: Wellness through Indian medicine, Concepts of Ayurveda, resources and conservation of Indian traditional medicine

Course Objectives:
1. To understand the concept of Traditional knowledge and its importance.
2. To be aware of the conventional framework of Indian Medical System.
3. To comprehend the therapeutic specialties of Ayurveda.
4. To familiarize the conservation strategies of medicinal plants.

Course Outcome:
After completing this course, the students will be able to:
i. Explain the scope and importance of traditional knowledge system of India
ii. Paraphrase the framework of Indian Medical System
iii. Understand the therapeutic approaches in Ayurveda
iv. Comprehend conservation strategies of medicinal plants.

Course syllabus

Unit I: Knowledge framework and history of Indian Medical System

Indian Medical Systems – Ayurveda, Unani, Siddha and Sowa Rigpa; Codified and non-codified medical knowledge in India; History of Development of Ayurveda in India&Ayurveda literature; Five element theory–Panchamahabutasiddhanta.

Unit II: Fundamental concepts of Ayurveda

Introduction to Ayurveda, Ashtanga Ayurveda (eight therapeutic specialities of Ayurveda);Three Doshas (Vata, Pitta and Kapha) the humors or body energies; Seven Dhatus, bodyissues, three major excretory functions and their importance; Prakriti: Body-mind types andits implication in healthcare; Three psychological states of mind (Gunas)Agni- Digestion andmetabolism; The daily and seasonal regimen (dinacharya and ritucharya) based on Ayurvedaprinciples.

Unit III: History and Present Botany in India

Introduction to the Plant Taxonomy; diversity of the plants in the country; Major forest typesof the India; Overview of important vegetation types across India; Conventional botany withits current multidisciplinary application; Important Botanist, their research and contrition for
the society.

Unit IV: Usage and Conservation of Medicinal Plants, Sacred groves

Medicinal plant wealth of India; Plants used in wellness; Spices in daily use; Wild edibleplants; Plants used in wellness in current life style pattern; Threatened medicinal plants ofIndia –threats to medicinal plant biodiversity; Conservation of medicinal plants - in situ andex-situ; Case studies; Sacred grove and their role in conservation; case study of the traditionalunderstanding and protection of Sacred groves (Case study).

Recommended Books:

1. Sastry K (ed.) Charaka Samhita-Hindi translation, part-1&2, Chowkhamba Orientalia Varanasi.
2. Sharma PV. History of medicine in India, Chowkhamba Orientalia Varanasi.
3. Vagbhata, Astanga Hrdayam-English by Prof. K.R. Srikantha Murthy, Prathama Khanda, Chowkhambakrishnadas academy, Varanasi; 2018.
4. Ved, D. K., S. Noorunnisa Begum and K. Ravikumar. Trade in Indian Medicinal Acharya Sushruta, Sushruta Samhita-Hindi by Dr.Anantaram Sharma, part-1; Acharya Priyavrita Sharma, Dravyaguna-Vijanana– Hindi, part-1, Chowkhamba bharti academy, Varanasi; 2020.
5. Goraya, G.S. & Ved, D.K. 2017. Medicinal Plants in India: An Assessment of their Demand and Supply. National medicinal Plants Board, Ministry of AYUSH, Government of India, New Delhi and Indian Council of Forestry Research Education, Dehradun.
6. Rajasekharan, P.E., Shabir Hussain Wani (Eds.). 2020. Conservation and Utilization of Threatened, Medicinal Plants, Springer International Publishing.
7. Sastry JLN, 2005. Dravyaguna Vijnana, Vol-I, Chaukhambha Orientalia, Varanasi.
8. Sastry JLN, 2005. Dravyaguna Vijnana, Vol-II, Chaukhambha Orientalia, Varanasi.
9. Peninsular India. Pullaiah, T., K.V. Krishnamurthy and Bir Bahadur (eds.) 2017. Apple Academic, Press, Oakville, Canada and Waretown, NJ. ISBN 978-1-77188-404-4.
10. Plants in Ethnobotany of India. Vol. 5. The Indo-Gangetic Region and Central India. Pullaiah, T., K.V. Krishnamurthy and Bir Bahadur (eds.)

Core Practical-II (Core 4+5+6)

4. Molecular Genetics

1. Setting of molecular biology laboratory creating of ribonuclease free environment in the laboratory.
2. Quantification of DNA and RNA by UV-spectrophotometer.
3. Isolation of total DNA from *E.coli* cells.
4. Isolation of total DNA from Plant leaf tissue
5. Isolation of total RNA from plant leaf tissue
6. Isolation of plasmid DNA from *E.coli* cells.
7. Agarose gel electrophoresis analysis of nucleic acids.

5. Concepts of Medical Genetics

1. Karyotyping
2. Assignments on genetic disorders and genetic counselling techniques
3. Blood group analysis: ABO typing, Rh(D) typing, and ABH secretor status
4. Genetic traits: Colour-blindness, Phenyl Thio Carbamide (PTC) sensitivity
5. Sickling test

6. Energy Metabolism

8. Isolation and estimation of glycogen/starch.
9. Determination of activity of α- Salivary amylase.
10. Isolation and assay of activity of acid phosphatase from potatoes.
11. Effect of substrate concentration on salivary amylase activity.
12. Effect of enzyme concentration on salivary amylase activity.
13. Effect of temperature on salivary amylase activity.
14. Effect of pH on salivary amylase activity.

Skill Practical II (Skill 3+4)

3. Genetic Engineering

1. Determination of type of nucleic acid by nucleases.
2. Restriction digestion of plasmid DNA.
3. Ligation of inset with Plasmid DNA.
4. Polymerase chain reaction (PCR).
5. Preparation of *E.coli* competent cells for transformation.
6. Transformation of Plasmid DNA into competent cells.
7. Screening and confirmation of recombinant clone.
8. Over-expression of recombinant proteins in *E. coli* system.
9. Purification and confirmation of recombinant proteins.
10. Sothern blotting (Capillary and diffusion methods).
11. Solving the black board problems.
12. Restriction digestion and comparison of methylated and unmethylated DNA by agarose gel.

4 .Concepts of Biostatistics and Population genetics

1. Collection, classification, and tabulation of biological data
2. Diagrammatic and graphical representation (bar, pie, histogram, polygon)
3. Calculation of central tendency (mean, median, mode)
4. Measures of dispersion (standard deviation, variance, coefficient of variation)
5. Analysis of skewness and kurtosis
6. Hypothesis testing with biological data
7. Student's *t*-test (one-sample and two-sample)
8. Chi-square test for Mendelian ratios (goodness of fit)
9. One-way and two-way ANOVA
10. Correlation and regression analysis
11. Calculation of standard error and confidence intervals
12. ABO and Rh(D) blood grouping
13. ABH secretor status testing
14. Sickling test for sickle cell trait/disease
15. Calculation of allele and genotype frequencies
16. Demonstration of Hardy-Weinberg equilibrium using simulations or data
17. Testing for color blindness using Ishihara charts
18. PTC (Phenyl Thio Carbamide) sensitivity test
19. Study of polygenic traits (e.g., height, skin color survey)
20. Karyotyping (human or plant metaphase chromosome analysis)
21. Estimation of heritability using simulated parent-offspring data

SEMESTER III

Core Course 7: Epigenetics

Course Objectives:
By the end of this course, students will:

1. Learn how DNA methylation, histone changes, and non-coding RNAs control genes
2. Examine how epigenetics affects diseases like cancer and immune disorders
3. Master techniques to study epigenetic modifications
4. Discover how epigenetics is used in treatments

Course Outcomes Summary:
Students will be able to:

1. Explain epigenetic mechanisms including DNA methylation, histone modifications, and non-coding RNA regulation
2. Demonstrate how epigenetic changes control gene expression through processes like X-inactivation and imprinting
3. Analyze the role of epigenetics in disease development, particularly cancer and genetic disorders
4. Evaluate epigenetic therapies and their applications in medicine
5. Apply various detection methods for DNA methylation, histone modifications, and non-coding RNA analysis
6. Connect epigenetic dysregulation to immune system function and autoimmune conditions

Course syllabus

UNIT-I: DNA METHYLATION AND HISTONE MODIFICATIONS

DNA methylation: Introduction, De Novo methylation, Maintaining methylation and DNA methylation and transcriptional silencing; DNA methylation in prokaryotes and eukaryotes; Histone modifications and Histone code- Acetylation, Methylation, Phosphorylation, Ubiquitination and ADP-Ribosylation and Sumoylation; Non-coding RNA (ncRNA)- MicroRNA(miRNA) biogenesis and function, small interfering RNA biogenesis and function and Epigenetic regulation by ncRNA.

UNIT-II: EPIGENETIC REGULATION OF GENE EXPRESSION

Epigenetic regulation of gene expression: Heterochromatin spreading and position effect variegation, transvection, paramutation, imprinting and X-chromosome inactivation; Epigenetics in Cancer: Epigenetic features of a normal cell, DNA hypomethylation in tumours, inactivation of tumor suppressor genes, histone modifications of cancer cells, role of miRNA in cancer, epigenetics in cancer management, epigenetic therapy of cancer.

UNIT-III: EPIGENETICS AND GENETIC SYNDROMES

Epigenetics and its genetic syndromes: Introduction, Chromatin remodeling- X-Linked Thalassemia Mental Retardation syndrome, CHARGE syndrome, Cockayne syndrome (CSB), ICF syndrome, Rett syndrome, CLS syndrome and FSHD. Epigenetics and Immunity: Introduction, Epigenetics in immune differentiation and

the immune response, Epigenetics in Autoimmunity, Epigenetic changes in other Autoimmune disorders.

UNIT-IV: EPIGENETIC DETECTION METHODS

DNA Methylation Detection Methods: Bisulfite conversion based : Whole-Genome Bisulfite Sequencing (WGBS), Methylation-Specific PCR (MSP),Pyrosequencing; Enzyme based: Methylation-Sensitive Restriction Enzymes (e.g., HpaII/MspI),Methylated DNA Immunoprecipitation (MeDIP); Histone Modification Detection Methods: Chromatin Immunoprecipitation Sequencing (ChIP-seq),CUT&Tag/CUT&RUN; Non-Coding RNA Analysis : RNA-Seq with Immunoprecipitation (RIP-seq/CLIP-seq).

Recommended Books:

1. Epigenomics by Anne C.Ferguson Smith(2009) .
2. Epigenetics and diseases by Susan M.Gasser, En Li(2011) .
3. Epigenetics in Biology and Medicine by NanelEsteller(2009).
4. Handbook of Epigenetics-The new molecular and medical genetics by Trygve Tollefsbol.
5. Epigenetics in Cancer-The new England journal of medicine by Manel Esteller(2008)
6. Advanced Molecular Biology by Twyman(1999).

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]

1. https://onlinecourses.nptel.ac.in/noc19_bt15/preview
2. https://www.classcentral.com/course/swayam-genetic-engineering-theory-and-application-14090
3. http://biology.kenyon.edu/courses/biol114/Chap08/Chapter_08a.html

Course Designed By: Dr. S. Selvakumar

Core Course 8: Concepts of Genomics

Course Objectives:

By the end of this course, students will:

1. Learn core genomics concepts, DNA markers, and gene mapping techniques.
2. Understand genetic mapping approaches (linkage/QTL/physical maps) and GWAS.
3. Master genome analysis tools (mutagenesis, TILLING, RNAi, CRISPR).
4. Study protein interaction methods (yeast two-hybrid, mass spectrometry, microarrays).

Course Outcomes:

Students will be able to:

1. Explain genomics approaches (forward/reverse genetics) and DNA markers
2. Design genetic maps and analyze QTL/heritability traits
3. Apply mutagenesis and genome-editing tools (CRISPR, RNAi)
4. Use protein interaction techniques for functional studies
5. Interpret genomic data for research/biotech applications

Course syllabus

UNIT-I: INTRODUCTION TO GENOMICS

Introduction to Genomics: Definitions, classification based on system attributes, relationships to other scientific disciplines and types of organisms studied. Historical perspective of Genomics; Concepts and applications of Functional genomics; Forward genetics and Reverse genetics approaches, Loss of function, Gain of function. DNA markers:-RFLP, AFLP, RAPD, SSRs, SNPs, CAPS, SCAR markers; Mapping in prokaryotes by Transformation, Mapping in prokaryotes by Transduction, Mapping in prokaryotes by Conjugation.

UNIT-II: GENETIC MAPPING TECHNIQUES

Construction of the genetic linkage maps:- human, plants; Development of F_2, BC and RIL mapping population; Map based cloning- Mutant Mapping, LOD score; Heritability and description of continuous variation of wheat kernel color and human skin color; QTL mapping.

Physical mapping: Cytogenetic maps of chromosomal banding, STS, FISH, restriction maps, radiation hybrid mapping (RH), clone contig maps; Genome wide association studies.

UNIT-III: GENOME ANALYSIS TOOLS

Mutagenesis as Functional Genomics Tool: T-DNA insertional mutagenesis, Transposon-based mutagenesis (Ac/Ds), Activation tagging, Floxing, Viral mediated transfection.Genome wide mutation screening: TILLING (Targeted Induced Local Lesion IN Genome) - principle and experimental approach, ECO-TILLING; DEALING (Detecting Adducts Local Lesion IN Genome); Site directed Mutagenesis.

UNIT-IV: GENE FUNCTION ANALYSIS TECHNIQUES

Gene functions through protein interactions: Identification of Protein–Ligand Interactions. Yeast Two-Hybrid Selection System: Analysis of genome-wide protein–protein interactions in organisms. Mass Spectrometry for Protein–Protein Interaction Mapping, DNA Microarray Technology- Types of Microarrays and Advantages, Experimental design; RNA silencing: Antisense RNA technology, RNAi and Si RNA; CRISPR (CRISPR/Cas9) - Mechanism and applications. Protein microarrays: overview, principle, limitations;

Recommended Books:

1. Protein Microarrays, edited by Mark schena, Jones and Bartlet publisher, 2005.
2. Microbial Functional Genomics, Jizhong Zhou, Dorothea K. Thompson, Ying Xu, James M. Tiedje, A John Wiley & Sons, Inc., Publication, 2004.
3. Microarrays for an Integratiul J. But. Kho and Atte, Published in India by Ane Books, 2003.
4. Gene Cloning and DNA analysis An Introduction, Sixth Edition, T. A. Brown, Wiley-Blackwell publications, A John Wiley & Sons, Inc., Publication, 2010.
5. Genomes 4th edition by T.A.Brown (2018)
6. 6.Principles of Gene Manipulation and genomics: An Introduction to genetic engineering by Primrose and Twyman, 7thEd. . 2013.
7. Microbial functional genomics by Jizhgahov, Dorothea K.Thompson Ying xu, James M.Tiedje.
8. 8.Protein Arrays, Biochips and Proteomics: The Next Phase of Genomic Discovery
9. by Joanna S. Albala and Ian Humphery-Smith,2019.

Core course 9: Immunology in Health and Disease

Course Objectives

By the end of this course, students will:

1. **Understand** innate and adaptive immunity, including immune cells (B/T/NK cells) and lymphoid organs.
2. **Master** antigen-antibody interactions and key techniques (ELISA, flow cytometry).
3. **Evaluate** immune responses in health and disease (hypersensitivity, autoimmunity, immunodeficiency).
4. **Design** strategies for vaccination and immunotherapy (e.g., HIV, cancer).

Course Outcomes

After completing this course, students should be able to:

1. **Explain** how innate and adaptive immune systems work together.
2. **Perform** serological tests (ELISA, agglutination) and interpret results.
3. **Analyze** complement pathways and their role in inflammation.
4. **Design** monoclonal antibodies for diagnostic/therapeutic use.
5. **Apply** WHO guidelines to vaccination programs

Course syllabus

UNIT-I: FOUNDATIONS OF IMMUNITY

Basic concepts in Immunology: Innate immunity and Adaptive immunity. Cells of the immune system - B cells, T cells, NK cells, phagocytes, inflammatory cells, antigen presenting cells, organs of immune system - primary, secondary and tertiary lymphoid organs. Immunohematology: blood groups, blood transfusion and Rh-incompatibility.

UNIT-II: ANTIGENS & ANTIBODY TECHNOLOGIES

Antigens - nature, types, factors influencing antigenecity, haptens, adjuvants and superantigens. Antibodies - structure, types, classes and functions. Antigen – antibody interactions: Immunological techniques: Flocculation, Precipitation, immunodiffusion, Phagocytosis, Opsonization, Enzyme Linked Immunosorbent Assay, Immunohistochemistry, Flow Cytometry. Polyclonal antibodies, Monoclonal antibodies: production and applications.

UNIT-III: IMMUNE RESPONSES & DISORDERS

Immune Response: Humoral immune response, cell mediated immune response. Recognition of antigen: MHC - Types, Antigen processing and presentation, activation and differentiation of B cells and T cells. Effector mechanisms: Cytokines, CTL, NK cell mechanism of cytolysis and Antibody dependent cell mediated

cytotoxicity. Complement activation pathways: Classical, alternate and lectin pathway. Hypersensitivity, Autoimmunity, Regulation of immune response. Vaccines, Immunization: Passive and active immunization, WHO recommended Immunization Schedule.

UNIT-IV: IMMUNOGENETICS & DISEASES

Immunogenetics: Organization, rearrangement and expression of Ig genes. Generation of antibody diversity, Inherited and acquired immunodeficiency diseases: Recessive gene defects, X- linked lymphoproliferative syndrom, SCID, Type 1 diabetes mellitus, multiple sclerosis, Inflammatory bowl disease, Rheumatoid arthritis, Chronic lymphocytic leukemia, haemophilia, sickle cell anemia, erythroblastosis fetalis, AIDS.

Recommended Books:

1. Essentials of Immunology - Ian Roitt - 13th Ed.Blackwell Scinentific Publications.2017.
2. Kuby Immunology by Jenni punt et al.,8th Ed.2018.
3.Paul's Fundamental Immunology by by Martin Flajnik et al.,2022.
4. Janeway's Immunobiology by Kenneth M. Murphy, Casey Weaver 9th Ed.,2016.

Related Online Contents [MOOC, SWAYAM, NPTEL, Websites etc.]
1. https://www.hindawi.com/journals/bmri/2014/437987/
2 . https://www.tandfonline.com/doi/full/10.1080/1744666X.2019.1623670
3.https://www.technologynetworks.com/immunology/articles/using-the-internet-of-things-tofight-virus-outbreaks-331992

Skill Oriented 5: Basics of*In Silico Genomics*

Course Objectives

By the end of this course, students will:

1. Understand the principles of *in silico* genomics and types of biological data (genomic DNA, cDNA, ESTs).
2. Navigate key databases (GenBank, UniProt, PDB) and retrieval systems (Entrez, SRS).
3. Analyze genomes using browsers (Ensembl, UCSC) and proteomics tools (ExPASy, STRING).
4. Perform sequence alignments (BLAST, Clustal Omega) and phylogenetic analysis (MEGA, PhyML).
5. Predict protein structures using modeling tools (SWISS-MODEL, AlphaFold2) and validate results (Ramachandran plots).

Course Outcomes

After completing this course, students should be able to:

1. **Explain** the significance of *in silico* genomics in modern biology.
2. **Retrieve** and deposit sequences in databases (GenBank, UniProt) using tools like BankIt.
3. **Compare** genomes using browsers (NCBI Map Viewer, KEGG) and interaction databases (BioGRID).
4. **Execute** sequence alignments (global/local/MSA) and interpret evolutionary relationships.
5. **Model** protein structures (homology/ab initio) and visualize them (PyMOL, Swiss-PDB Viewer).
6. **Validate** predicted structures using computational tools (MolProbity, ProCheck).

Course syllabus

UNIT-I: INTRODUCTION TO IN-SILICO GENOMICS

Introduction to In-Silicogenomics and its significance in the current synario; Types of Biological data:- Genomic DNA, Complementary DNA, Recombinant DNA, Expressed sequence tags, Sequence-Tagged Sites; Data bases:-Sequence databases: Nucleic acid sequence databases: GenBank, EMBL, DDBJ; Protein sequence databases: Uniprot-KB: SWISS-PROT, TrEMBL, UniParc; Structure Databases: PDB, PDBsum, NDB, PubChem, ChemBank. Literature Databases: - Open access and open sources, PubMed, PLoS, Biomed Central, NAR databases; Sequence file formats: GenBank, FASTA, GCG, MSF etc. Database Retrieval and Deposition Systems: SRS, Entrez, Bankit, Seqin, Webin.

UNIT-II: GENOME AND PROTEOME DATABASES

Genome Databases: Viral genome database: -ICTVdb; Bacterial Genomes database: - Ensembl Bacteria, Microbial Genome Database-MBGD; Genome Browsers: - Ensembl, VEGA genome browser, NCBI-NCBI map viewer, KEGG, MIPS, UCSC

Genome Browser; Gene Expression databases: - GEO, SAGE. Protein-Protein interaction databases: - STRING, BioGRID, MINT; Proteomics tools: - ExPASy server, EMBOSS.

UNIT-III: SEQUENCE ANALYSIS AND PHYLOGENETICS

Sequence Comparison: Introduction to sequence alignment methods (global alignment, local alignment); Pairwise sequence alignment; Multiple sequence alignment (MSA). Applications of sequence alignment in identifying homologous sequences, detecting conserved motifs, and evolutionary analysis. sequence comparison tools: BLAST, Clustal Omega.
Phylogeny: Basic concepts, Terminology of phylogenetic tree: Root, Branch, Node, Leaf, Clade, Outgroup, Homology, Orthology, Paralogy, Xenology, Different types of trees- Rooted vs. Unrooted trees, Tools for phylogenetic analysis (PhyML, MEGA).

UNIT-IV: PROTEIN STRUCTURE PREDICTION

Protein structure prediction: Introduction to protein structure determination methods (X-ray crystallography, Cryo-EM, NMR spectroscopy). Secondary structural elements – Structural elements: backbone, domains, side chains, native state/conformation, Backbone flexibility- Φ and ψ, α and β propensities, Ramachandran plot, Prediction methods: Chou-Fasman Method & GOR Method; Tertiary structure prediction techniques: including homology modeling, threading (fold recognition), and ab initio methods. Protein structure validation methods (Ramachandran Plot analysis, MolProbity, Procheck). Tools and software for structure prediction and modeling (SWISS-MODEL, Alplhafold2). Visualization tools: RasMol, Swiss PDB viewer, Pymol.

Recommended Books:

1. Lesk, A.M. 2005, 2nd edition, Introduction to Bioinformatics. Oxford University Press.
2. Bioinformatics-A practical guide to the analysis of Genes and proteins, by Andreas D.Baxevanis, B.F.Francis Ouellette,3rd.
3. Bioinformatics-Tools and applications by David Edwards, Jason strajich and David Hansen(2009)
4. Fundamental Concepts of Bioinformatics - Dan E. Krane, Michael L. Raymer, Pearson education (2004)

Skill Oriented Course 6: Pathogenomics

Course Objectives

By the end of this course, students will:

1. Understand epidemiological principles and microbial pathogenesis mechanisms.
2. Analyze emerging/re-emerging diseases (COVID-19, HIV, TB, HPV) through genomic lenses.
3. Apply advanced molecular diagnostics (PCR, microarrays) and bioinformatics tools.
4. Design genomic strategies for novel antibiotics and vaccines (reverse vaccinology).
5. Evaluate host-pathogen interactions and public health interventions.

Course Outcomes:

Students will be able to:

1. Analyze Disease Transmission & Host-Pathogen Interactions
2. Apply Genomic Tools for Diagnosis
3. Decipher Pathogen Genomics
4. Design Therapeutic Strategies
5. Innovate Vaccines via Genomics

Course syllabus

UNIT-I: PRINCIPLES OF EPIDEMIOLOGY AND DIAGNOSTICS

Epidemiology: Principles of epidemiology, Pandemic diseases, Common epidemic diseases in humans. Agent, sources and reservoir, Stages of disease progression, infectious disease transmission- Direct host –to-host, indirect host-to-host transmission; Natural Host resistance, Harmful microbial interactions with humans: entry, colonization and growth, virulence, virulence factors and toxins; Host risk factors in infection: age, stress and diet, compromised host. New diagnostics Advanced molecular diagnostic methods for detection of microbial infections: PCR, SNP, FISH, Flow-cytometry, Microarray.

UNIT-II: EMERGING DISEASES AND HIV THERAPIES

Emerging and reemerging infectious diseases- Chikungunya, Swine flu (H1N1), Dengue, Covid-19 -history, symptoms, viral replication, genetic variability, diagnosis, prevention.

HIV pathogenesis-gene-therapy: Overview of HIV pathogenesis: structure and genome, replication, target-cell depletion and apoptosis, genetic variability; Bioinformatic analysis of HIV. Ribozyme as gene therapeutic agents for HIV/AIDS: Ribozyme design and in vitro efficacy.

UNIT-III: TUBERCULOSIS AND HPV GENOMICS

Genomics of the Mycobacterium tuberculosis and BCG vaccines: Mycobacterium pathogenesis- tuberculosis, drug resistance, information from complete genome sequencing of Mycobacterium tuberculosis, strain-to –strain variability with M.tuberculosis sp., genomic analysis of M.bovis BCG vaccines;

HPV pathogenesis-microarray technology: Application of microarray technology in understanding HPV pathogenesis, Tissue-culture systems for studying HPV, alternation of cellular gene expression during latent infection by HPV.

UNIT-IV: APPLIED PATHOGENOMICS

Applications of Patho Genomics: Search for new antibiotics, Need for novel antibiotics, genomic technologies in antibacterial research, targeting the resistance mechanism, extremely narrow-spectrum drugs, strategies for reducing virulence, gene therapy; Recombinant Vaccines: Polyvalent vaccine, Subunit Vaccines, DNA Vaccines. Reverse vaccinology: Development of a vaccine against meningococcus B (MenB) by reverse vaccinology

Recommended Books:

1. Human Genetics by Gardner and Davies, 2nd Ed. 2012
2. Elements of Medical Genetics, Emery's. 15th Ed. 2017
3. Medical genetics / Lynn B. Jorde, John C. Carey, Michael J. Bamshad. 6th Ed. 2020
4. Human Genetics by Gangane, 6th Ed. 2021.
5. Principles of Genetics, by Snustad. Simmons. 7th Ed. 2015.

Core Practical-III (Core 7+8+9)

7. Epigenetics

1. DNA Methylation Analysis
   (Extract DNA and treat with methylation-sensitive enzymes (*HpaII* vs. *MspI*), then run gel electrophoresis.Compare banding patterns to infer methylation status.)
2. Chromatin Staining
   (Stain onion root tips with aceto-orcein/DAPI to observe condensed vs. relaxed chromatin under a microscope.Compare treated (e.g., with TSA, a deacetylase inhibitor) vs. untreated samples.)
3. Environmental Epigenetics
   (Grow *Arabidopsis* or *Drosophila* under stress (salt/UV) and check for chromatin changes via staining.)
4. Bisulfite Conversion Demo
   (Simulate bisulfite treatment on plasmid DNA, PCR amplify, and run gel to see cytosine-to-uracil conversion.)
5. **Public Methylation Data Analysis**
   (Use **UCSC Genome Browser** or **MethBank** to visualize CpG island methylation in different tissues.)
6. **ChIP-Seq Data Exploration**
   (Load public **ENCODE ChIP-seq data** into **IGV** to study histone modifications (e.g., H3K27ac).
7. **Methylation-Gene Expression Correlation**
   (Download RNA-seq & methylation data from **GEO**, analyze in *R* (methylKit) to find inverse correlations.)

8. Concepts of Genomics

1. To study genetic polymorphism in plant samples using RAPD
2. Retrieving Gene Expression Data from Public Microarray Genomics Databases: **GEO (Gene Expression Omnibus),ArrayExpress, Stanford Microarray Database (SMD)***(now archived),* **Oncomine, CIBEX (Center for Information Biology Gene Expression Database)***(Archived),* **The Cancer Genome Atlas (TCGA),Expression Atlas (EMBL-EBI)**
3. To retrieve and compare gene expression levels between cancerous and normal tissue samples using public microarray databases such as GEO, TCGA, and Oncomine.
4. Compare gene expression profiles between drought-stressed and healthy (control) plants using public microarray databases.
5. Retrieve and compare gene expression levels between diseased and healthy plant tissue samples using publicly available microarray datasets (e.g., from GEO or ArrayExpress).
6. Retrieve and compare gene expression levels between cancerous and normal tissue samples using public microarray databases such as GEO, TCGA, and Oncomine.
7. Map and analyze protein-protein interactions (PPIs) for a disease-related gene (e.g., *BRCA1 or TP53*) using the STRING database.

9. Immunology in Health and disease

1. Preparation of serum and plasma from blood.
2. Trypan blue exclusion test to measure the viability of the cells.
3. Isolation of peripheral blood lymphocytes by Ficoll- Hypaque gradient.
4. Mixed Lymphocyte Reaction assay.
5. MTT assay to measure the proliferation of lymphocytes.
6. Cell counting by Hemocytometer (WBC and RBC).
7. Quantitative precipitation test:
   (A) Redial immunodiffusion.
   (B) Ouchterloney double diffusion.
8. Immuno-elecrophoresis
9. Venereal Disease Research Laboratory (VDRL) test for syphilis
10. Widal agglutination test for typhoid fever
11. Determination of A, B, O and Rh grouping & Rh typing by Agglutination.
12. Hepatitis B surface antigen test
13. Human Chorionic Gonadotropin (HCG) test
14. Enzyme Linked Immunosorbent Assay (ELISA) / Tridot test.

Skill Practical III (Skill 5+6)

5. Basics of In-Silico genomics

1. Retrieve nucleotide sequences from GenBank using NCBI Entrez.
2. Download and interpret a gene sequence in FASTA and GenBank formats.
3. Search and explore scientific articles related to genomics using PubMed.
4. Submit a nucleotide sequence to GenBank using BankIt (demo-based activity).
5. Explore gene expression datasets related to plant/cancer studies using GEO.
6. Visualize a bacterial genome using Ensembl Bacteria genome browser.
7. Analyze protein-protein interactions using the STRING database.
8. Retrieve and analyze a protein sequence from UniProt using ExPASy tools.
9. Perform pairwise sequence alignment using BLAST (nucleotide/protein).
10. Conduct multiple sequence alignment using Clustal Omega and identify conserved regions.
11. Construct and visualize a phylogenetic tree using MEGA software.
12. Predict the tertiary structure of a protein using SWISS-MODEL and visualize it using PyMOL.
13. Designing of primers for gene amplification

6. Pathogenomics

1. Identify direct vs. indirect modes of disease transmission using case-based scenarios.
2. Detection of diseases by using immuno and molecular tools.
3. Genomics sequencing and detection of pathogens
4. Prediction of B and T cell epitopes from protein sequence of pathogens.
5. Analysis of epitopes for toxicity, allergenicity and antigenicity.
6. Epitope and receptor structure prediction and validation.

SEMESTER IV

OpenOnlineSkill Development Course/s

(Students can choose an8-credit course or 4 Course from the courses available in the ONLINE SWAYAM&SWAYAM PLUS)
https://swayam.gov.in/
https://swayam-plus.swayam2.ac.in/
https://onlinecourses.swayam2.ac.in/ntr25_ed27/preview

Project Work

APPENDEX-I

MARKS DISTRIBUTION FOR THEORY AND PRACTICAL EXAMINATIONS

SEMESTER-I	**Total Marks**	**Summative Examination (60 Marks)**	**FormativeInternalAssessment(40marks)**			
		External	**Internal**	**Attendance**	**Quiz**	**Students Seminar**
CORE-1	100	60	25	5	5	5
CORE-2	100	60	25	5	5	5
CORE-3	100	60	25	5	5	5
SKILL-1	100	60	25	5	5	5
SKILL-2	100	60	25	5	5	5
CORE PRACTICAL-I (CORE 1+2+3)	150					
SKILL PRACTICAL-I (SKILL 1+2)	100					
TOTAL MARKS	**750**					
SEMESTER-II	**Total Marks**	**Summative Examination (60 Marks)**	**FormativeInternalAssessment(40marks)**			
		External	**Internal**	**Attendance**	**Quiz**	**Students Seminar**
CORE-4	100	60	25	5	5	5
CORE-5	100	60	25	5	5	5
CORE-6	100	60	25	5	5	5
SKILL-3	100	60	25	5	5	5
SKILL-4	100	60	25	5	5	5
CORE PRACTICAL-II (CORE IV+V+VI)	150					
SKILL PRACTICAL-II (SKILL 3+4)	100					
TOTAL MARKS	**750**					

		Summative Examination (60 Marks)	FormativeInternalAssessment(40marks)			
SEMESTER-III	**Total Marks**	**External**	**Internal**	**Attendance**	**Quiz**	**Students Seminar**
CORE-7	100	60	25	5	5	5
CORE-8	100	60	25	5	5	5
CORE-9	100	60	25	5	5	5
SKILL-5	100	60	25	5	5	5
SKILL-6	100	60	25	5	5	5
CORE PRACTICAL-III (CORE 7+8+9)	150					
SKILL PRACTICAL-III (SKILL 5+6)	100					
TOTAL MARKS	**750**					
SEMESTER-IV	**Total Marks**					
PROJECT WORK	350	Dissertation 200	Presentation 50 +50+50 **(Viva Voce Continuous Internal assessment**			
TOTAL MARKS	**2600**					

1. Each CORE or SKILL Theory Paper consists of FOUR (4) Units. Theory examination is for 100 Marks (60 Marks external +25 Marks internal examination +05 Marks Attendance +05 Marks Quiz + 05 Marks Students Seminar).

2. Each Practical Paper is for 150 Marks. For the Semester End External Practical examination, THREE CORE (03) Practical's (50+50+50) may be combined, and the examination will be conducted for 150 Marks for FOUR (04) hours. TWO Skill-Oriented Courses (50+50) may be combined, and an examination will be conducted for 100 Marks for FOUR (04) Hours.

MODEL QUESTION PAPER

I SEMESTER
PAPER CODE AND TITLE:
(w.e.f. Academic Year 2025-2026)
(No additional sheet will be supplied)

Time: 3 hours **Max. Marks: 60**

PART – A (Marks: **4x3= 12 Marks**)
(Two questions from each Unit Answer any **FOUR**questions.
Each answer should not exceed ONE (1) Page
All questions carry equal marks

1. From **UNIT - I**
2. From **UNIT - I**
3. From **UNIT - II**
4. From **UNIT - II**
5. From **UNIT - III**
6. From **UNIT - III**
7. From **UNIT - IV**
8. From **UNIT - IV**

PART B (Marks: **4 x 12 = 48**)
Answer **ALL** questions.
Two questions from each Unit with internal choice, answer should not exceed SIX (6) Pages
All questions carry equal marks.

UNIT – I

9.

OR

10.

UNIT – II

11.

OR

12.

UNIT – III

13.

OR

14.

UNIT – IV

15.

OR